Calculating Concentration

Exam code: 0620 & 0971

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Units of concentration

- A **solute** is a solid substance that dissolves into a liquid
  - The amount of solute can be expressed in **grams (g)** or **moles (mol)**
- A **solvent** is the liquid that a solute dissolves in
  - The amount / volume of a solvent is measured in **cm^3** or **dm^3**
- Most chemical reactions occur between solutes which are dissolved in solvents, such as water or an organic solvent
- A **solution** is the mixture formed when a solute dissolves in a solvent
  - The amount / volume of a solution measured in **cm^3** or **dm^3**
- **Concentration** refers to the amount of solute there is in a specific volume of the solvent
  - The greater the amount of solute in a given volume, the greater the concentration
  - Concentration is sometimes commonly referred to as strength
    - For example, dissolving more coffee in hot water results in a stronger coffee
- Typically, concentration is expressed in terms of the amount of substance per dm^3
  - Therefore, the units of concentration are:
    - **g / dm^3**
    - **mol / dm^3**

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Calculating concentration

Extended tier only

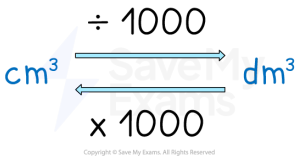
Calculating concentration using mass

- The general formula to calculate the concentration in g / dm^3 is:

$$\text{concentration (g / dm}^3\text{)} = \frac{\text{mass of solute (g)}}{\text{volume of solution (dm}^3\text{)}}$$

- Concentration can also be measured in grams per cubic decimetre
  - 1 decimetre cubed (dm^3) = 1000 cm^3
  - 1 decimetre cubed (dm^3) is the same as 1 litre
- You may be given data in a question which needs to be converted from cm^3 to dm^3 or the other way around

Converting cm^3 and dm^3



To go from cm^3 to dm^3 divide by 1000. To go from dm^3 to cm^3 multiply by 1000

Worked Example

A student dissolved 10 g of sodium hydroxide, NaOH, in 2 dm^3 of distilled water. Calculate the concentration of the solution in g / dm^3.

Answer:

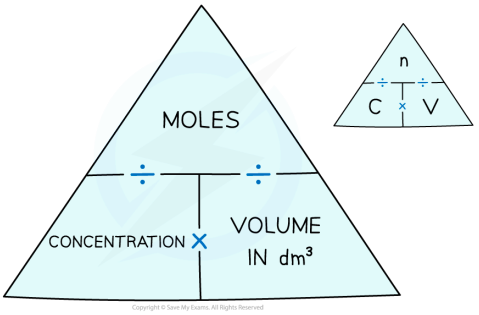
- Write down the information you are given in the question:
  - Mass of solute: 10 g
  - Volume of solution: 2 dm^3
- Calculate the concentration:
  - Concentration = $\frac{\text{mass of solute (g)}}{\text{volume (dm}^3\text{)}}$
  - Concentration = $\frac{10\text{ g}}{2\text{ dm}^3}$ = **5 g / dm^3**

Calculating concentration using moles

- Concentration can therefore be expressed in moles per decimetre cubed and calculated using the following equation:

$$\text{concentration (mol / dm}^3\text{)} = \frac{\text{number of moles of solute (mol)}}{\text{volume of solution (dm}^3\text{)}}$$

- We can modify the concentration formula to include moles
  - The units in the answer can be written as mol / dm^3 (this can also be written as mol dm^{-3})
- You may have to convert from g / dm^3 into mol / dm^3 and vice versa depending on the question
- To go from g / dm^3 to mol / dm^3
  - Divide by the molar mass in grams
- To go from mol / dm^3 to g / dm^3
  - Multiply by the molar mass in grams
- Some students find formula triangles help them to understand the relationship:



The concentration-moles formula triangle can help you solve these problems

Worked Example

Calculate the amount of solute, in moles, present in 2.5 dm^3 of a solution whose concentration is 0.2 mol / dm^3.

Answer:

- Write down the information you are given in the question:
  - Concentration of solution: 0.2 mol / dm^3
  - Volume of solution: 2.5 dm^3
- Calculate the number of moles:
  - Moles = concentration x volume
  - Moles = 0.2 x 2.5 = **0.5 mol**

Worked Example

Relative atomic masses, A_r: Na = 23; H = 1; O = 16

Answer:

- Calculate the M_r of NaOH:
  - 23 + 16 + 1 = 40
- Determine the number of moles of NaOH:
  - 40 g = 1 mole
  - So, 80 g = 2 moles
- Convert cm^3 to dm^3:
  - $\frac{500}{1000} = 0.5\ dm^3$
- Calculate the concentration:
  - Concentration = $\frac{moles}{volume}$
  - Concentration = $\frac{2}{0.5} = 4\ mol / dm^3$

Worked Example

25.0 cm^3 of 0.050 mol / dm^3 sodium carbonate was completely neutralised by 20.00 cm^3 of dilute hydrochloric acid. Calculate the concentration in mol / dm^3 of the hydrochloric acid.

$$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$$

Answer:

- Calculate the moles of sodium carbonate:
  - Moles of Na_2CO_3 = concentration x volume
    - **Remember:** The volume needs to be in dm^3
  - Moles of $Na_2CO_3 = 0.05 \times \frac{25.0}{1000} = 0.00125$
- Calculate the moles of hydrochloric acid:
  - The balanced symbol equation shows that 1 mole of Na_2CO_3 reacts with 2 moles of HCl
  - So, 0.00125 moles of Na_2CO_3 reacts with 0.00250 moles of HCl
- Calculate the concentration of hydrochloric acid:
  - Concentration = $\frac{moles}{volume}$
    - **Remember:** The volume needs to be in dm^3
    - 20 cm^3 ÷ 1000 = 0.02 dm^3
  - Concentration = $\frac{0.00250}{0.02} = 0.125\ mol / dm^3$

Examiner Tips and Tricks

Remember: Always convert the units from cm^3 to dm^3 by dividing by 1000.

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